

Electronic Nose Project

Beef Quality Classification

(c)2024 Djoko Purwanto, Dept. of Electrical Engineering Institut Teknologi Sepuluh Nopember (ITS) djoko@its.ac.id

```
student_name = 'Muhammad Zahran Razzaq'           # Write your name here
print(f'Hello {student_name} !')
```

Hello Muhammad Zahran Razzaq !

```
import platform
import psutil
import tensorflow as tf
from datetime import datetime

def get_system_info():
    info = {}
    # Get system information
    info['platform'] = platform.system()
    info['platform-version'] = platform.version()
    info['architecture'] = platform.machine()
    info['hostname'] = platform.node()
    info['processor'] = platform.processor()
    info['ram'] = str(round(psutil.virtual_memory().total / (1024.0
**3))) + ' GB'
    return info

print(f'Program start ...')
# Print the system information
system_info = get_system_info()
for key, value in system_info.items():
    print(f'{key}: {value}')

# Print TensorFlow version
print(f'TensorFlow version: {tf.__version__}')
if tf.config.list_physical_devices('GPU'):
    print(f'GPU is being used by TensorFlow')
else: print(f'GPU is unavailable for TensorFlow')

# Get the current time
current_time = datetime.now().strftime('%Y-%m-%d %H:%M:%S')
# Print the current time
print(f'The current time is: {current_time}')
```

```

Program start ...
platform: Windows
platform-version: 10.0.26200
architecture: AMD64
hostname: LAPTOP-9NNH1THV
processor: Intel64 Family 6 Model 183 Stepping 1, GenuineIntel
ram: 20 GB
TensorFlow version: 2.21.0
WARNING:tensorflow:TensorFlow GPU support is not available on native
Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU
will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
GPU is unavailable for TensorFlow
The current time is: 2026-09-28 10:28:32

```

(1) Read and Visualize Data

```

import pandas as pd

df_sheet = pd.read_excel('e-nose_dataset_12_beef_cuts.xlsx',
sheet_name='2.Round')
df_sheet

```

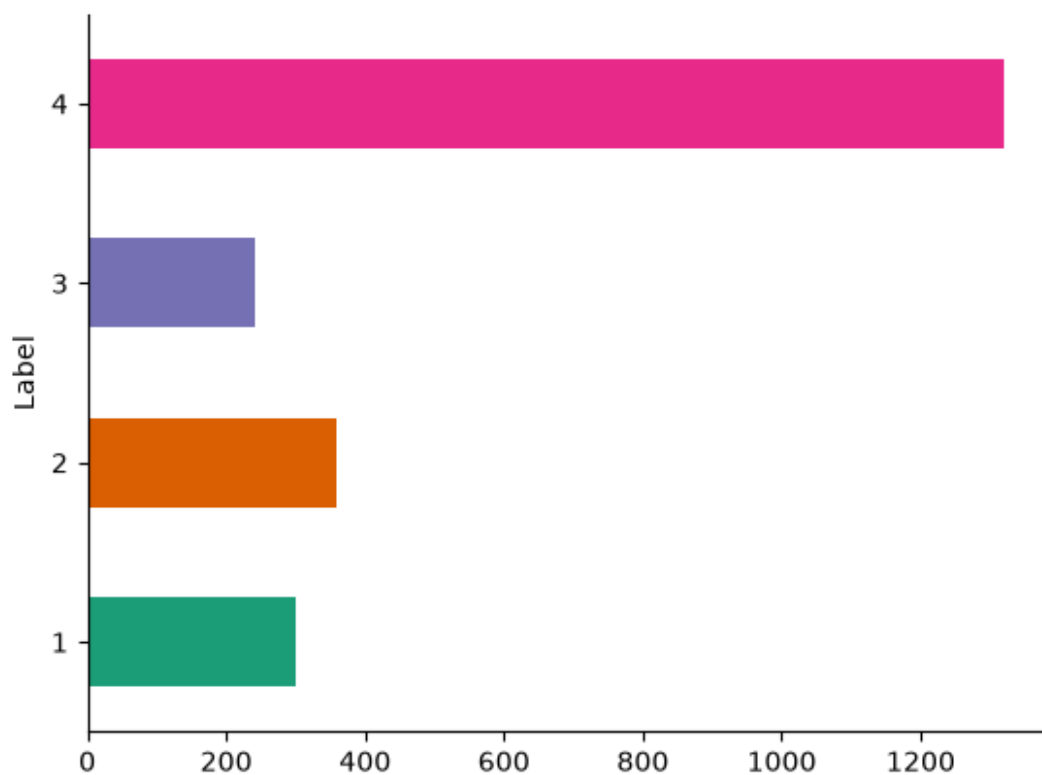
	Minute	TVC	Label	MQ135	MQ136	MQ137	MQ138	MQ2
MQ3 \								
0	1	1.875890	1	14.87	37.65	5.16	20.37	12.25
16.24								
1	2	1.875890	1	14.94	38.56	5.09	20.46	13.37
16.38								
2	3	1.875890	1	15.00	39.50	5.07	20.46	12.40
17.01								
3	4	1.875890	1	14.87	40.48	5.05	20.37	12.50
16.44								
4	5	1.875890	1	14.87	40.73	5.00	20.27	12.01
16.38								
...	...	...	...	...	...	...	...	...
.								
2215	2216	5.758033	4	11.48	3.15	27.25	13.06	10.97
8.70								
2216	2217	5.758033	4	11.35	3.12	27.53	12.90	10.93
8.53								
2217	2218	5.758033	4	11.26	3.14	27.67	12.90	10.68
8.63								
2218	2219	5.758033	4	11.35	3.12	27.39	12.90	10.35
8.60								
2219	2220	5.758033	4	11.26	3.14	27.67	12.85	10.55
8.63								
	MQ4	MQ5	MQ6	MQ8	MQ9			
0	28.66	6.11	3.59	46.19	10.01			
1	28.50	5.74	3.39	46.19	10.05			

2	28.18	6.18	3.25	46.83	10.21
3	28.18	5.60	3.16	46.19	10.13
4	28.50	5.48	3.11	47.48	10.13
...	...	...	...	...	...
2215	3.01	17.67	34.93	33.36	8.97
2216	2.96	16.93	34.47	33.55	8.97
2217	2.96	17.00	36.62	37.90	9.01
2218	2.95	17.14	34.70	34.51	9.04
2219	2.96	17.44	34.25	33.74	9.11

[2220 rows x 14 columns]

```
from matplotlib import pyplot as plt
import seaborn as sns

df_sheet.groupby('Label').size().plot(kind='barh',
color=sns.palettes.mpl_palette('Dark2'))
plt.gca().spines[['top', 'right']].set_visible(False)
```



(2) Create Dataset

```
df_dataset = df_sheet.copy()

encode_data = pd.get_dummies(df_dataset.Label)
df_dataset = pd.concat([df_dataset, encode_data], axis=1)
```

```
df_dataset.drop(columns=['Minute', 'TVC', 'Label'], inplace=True)
df_dataset
```

	MQ135	MQ136	MQ137	MQ138	MQ2	MQ3	MQ4	MQ5	MQ6
MQ8 \									
0	14.87	37.65	5.16	20.37	12.25	16.24	28.66	6.11	3.59
46.19									
1	14.94	38.56	5.09	20.46	13.37	16.38	28.50	5.74	3.39
46.19									
2	15.00	39.50	5.07	20.46	12.40	17.01	28.18	6.18	3.25
46.83									
3	14.87	40.48	5.05	20.37	12.50	16.44	28.18	5.60	3.16
46.19									
4	14.87	40.73	5.00	20.27	12.01	16.38	28.50	5.48	3.11
47.48									
...	...	...	...	...	...	...	...	...	...
...									
2215	11.48	3.15	27.25	13.06	10.97	8.70	3.01	17.67	34.93
33.36									
2216	11.35	3.12	27.53	12.90	10.93	8.53	2.96	16.93	34.47
33.55									
2217	11.26	3.14	27.67	12.90	10.68	8.63	2.96	17.00	36.62
37.90									
2218	11.35	3.12	27.39	12.90	10.35	8.60	2.95	17.14	34.70
34.51									
2219	11.26	3.14	27.67	12.85	10.55	8.63	2.96	17.44	34.25
33.74									

	MQ9	1	2	3	4
0	10.01	True	False	False	False
1	10.05	True	False	False	False
2	10.21	True	False	False	False
3	10.13	True	False	False	False
4	10.13	True	False	False	False
...	...	...	...	...	...
2215	8.97	False	False	False	True
2216	8.97	False	False	False	True
2217	9.01	False	False	False	True
2218	9.04	False	False	False	True
2219	9.11	False	False	False	True

[2220 rows x 15 columns]

```
import numpy as np
```

```
ds_values = df_dataset.values
```

```
ds_inputs = ds_values[:,0:11]
```

```
ds_outputs = ds_values[:,11:]
```

```

print(f'input shape : {ds_inputs.shape}')
print(f'output shape : {ds_outputs.shape}')
print(ds_inputs[0:5,:])
print(ds_outputs[0:5,:])

input shape : (2220, 11)
output shape : (2220, 4)
[[14.87 37.65 5.16 20.37 12.25 16.24 28.66 6.11 3.59 46.19 10.01]
 [14.94 38.56 5.09 20.46 13.37 16.38 28.5 5.74 3.39 46.19 10.05]
 [15.0 39.5 5.07 20.46 12.4 17.01 28.18 6.18 3.25 46.83 10.21]
 [14.87 40.48 5.05 20.37 12.5 16.44 28.18 5.6 3.16 46.19 10.13]
 [14.87 40.73 5.0 20.27 12.01 16.38 28.5 5.48 3.11 47.48 10.13]]
[[True False False False]
 [True False False False]
 [True False False False]
 [True False False False]
 [True False False False]]

```

Input normalization

```

inp_min = np.min(ds_inputs); inp_max = np.max(ds_inputs)
print(f'input --> min = {inp_min}; max = {inp_max}')

input --> min = 2.4; max = 54.08

```

Set new input min and input max, then do input normalization

```

def normalization(x,xmin,xmax):
    return 2*(x-xmin)/(xmax-xmin) - 1

inp_min, inp_max = 0.0, 60.0
ds_inputs_norm = normalization(ds_inputs, inp_min, inp_max)

print(f'input normalization shape: {ds_inputs_norm.shape}')
print(ds_inputs_norm[0:5,:])

input normalization shape: (2220, 11)
[[-0.5043333333333333 0.2549999999999999 -0.828 -0.32099999999999995
 -0.5916666666666667 -0.4586666666666667 -0.04466666666666663
 -0.7963333333333333 -0.8803333333333333 0.5396666666666665
 -0.6663333333333333]
 [-0.502 0.2853333333333334 -0.8303333333333334 -0.31799999999999995
 -0.5543333333333333 -0.45400000000000007 -0.050000000000000044
 -0.8086666666666666 -0.887 0.5396666666666665 -0.665]
 [-0.5 0.31666666666666665 -0.831 -0.31799999999999995
 -0.5866666666666667 -0.43299999999999994 -0.06066666666666665 -0.794
 -0.8916666666666666 0.5609999999999999 -0.6596666666666666]
 [-0.5043333333333333 0.3493333333333333 -0.8316666666666667
 -0.32099999999999995 -0.5833333333333333 -0.45199999999999996
 -0.06066666666666665 -0.8133333333333334 -0.8946666666666667

```

```

0.5396666666666665 -0.6623333333333333]
[-0.5043333333333333 0.3576666666666666 -0.8333333333333334
-0.32433333333333336 -0.5996666666666667 -0.45400000000000007
-0.0500000000000000044 -0.8173333333333334 -0.8963333333333333
0.5826666666666667 -0.6623333333333333]]

```

Split dataset into 80% train and 20% test set

```

from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(ds_inputs_norm,
ds_outputs, test_size=0.2, random_state=42)

print(f'Train set --> input shape: {X_train.shape}, output shape:
{Y_train.shape}')
print(f'Test set --> input shape: {X_test.shape}, output shape:
{Y_test.shape}')

Train set --> input shape: (1776, 11), output shape: (1776, 4)
Test set --> input shape: (444, 11), output shape: (444, 4)

```

(3) Build Model

```

model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(20, activation='sigmoid',
input_dim=X_train.shape[1]),
    tf.keras.layers.Dense(4, activation='softmax') ])

model.summary()
print('Model created')

c:\Users\Zahran\miniconda3\envs\dsec\Lib\site-packages\keras\src\
layers\core\dense.py:107: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in
the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

```

Model: "sequential"

Layer (type)	Output Shape
Param #	
dense (Dense)	(None, 20)
240	

dense_1 (Dense)	(None, 4)
-----------------	-----------

84 |

Total params: 324 (1.27 KB)
Trainable params: 324 (1.27 KB)
Non-trainable params: 0 (0.00 B)
Model created

(4) Train model

```
import time
import datetime

epochs = 5000
batch_size = int(0.1*len(X_train))
print('Batch size: ',batch_size)

X_train_tensor = tf.convert_to_tensor(X_train, dtype=tf.float32)
X_test_tensor = tf.convert_to_tensor(X_test, dtype=tf.float32)
Y_train_tensor = tf.convert_to_tensor(Y_train, dtype=tf.int32)
Y_test_tensor = tf.convert_to_tensor(Y_test, dtype=tf.int32)

reduce_lr = tf.keras.callbacks.ReduceLROnPlateau(
    monitor='val_accuracy', factor=0.1, patience=50, min_lr=1e-6)
early_stopping = tf.keras.callbacks.EarlyStopping(
    monitor='accuracy', patience=75, restore_best_weights=True)

optimizer = tf.keras.optimizers.Adam(learning_rate = 1e-2)
model.compile(loss='categorical_crossentropy', optimizer=optimizer,
metrics=['accuracy'])

starttime = time.time()

model_history = model.fit(X_train_tensor,Y_train_tensor,
    epochs=epochs,
    validation_data=(X_test_tensor, Y_test_tensor), verbose=2,
    batch_size=batch_size,
    callbacks=[reduce_lr, early_stopping] )

stoptime = time.time()-starttime
trainingtime = datetime.timedelta(seconds=stoptime)
print(f'\nTraining time : {trainingtime}')
print(f'End of training')

Batch size: 177
Epoch 1/5000
```

11/11 - 1s - 85ms/step - accuracy: 0.3164 - loss: 1.4697 -
val_accuracy: 0.5833 - val_loss: 1.1437 - learning_rate: 0.0100
Epoch 2/5000
11/11 - 0s - 11ms/step - accuracy: 0.5974 - loss: 1.1046 -
val_accuracy: 0.5833 - val_loss: 1.0782 - learning_rate: 0.0100
Epoch 3/5000
11/11 - 0s - 9ms/step - accuracy: 0.5974 - loss: 1.0306 -
val_accuracy: 0.5833 - val_loss: 0.9683 - learning_rate: 0.0100
Epoch 4/5000
11/11 - 0s - 9ms/step - accuracy: 0.5974 - loss: 0.9374 -
val_accuracy: 0.5833 - val_loss: 0.8910 - learning_rate: 0.0100
Epoch 5/5000
11/11 - 0s - 9ms/step - accuracy: 0.5985 - loss: 0.8475 -
val_accuracy: 0.6216 - val_loss: 0.7802 - learning_rate: 0.0100
Epoch 6/5000
11/11 - 0s - 8ms/step - accuracy: 0.6729 - loss: 0.7490 -
val_accuracy: 0.7027 - val_loss: 0.6816 - learning_rate: 0.0100
Epoch 7/5000
11/11 - 0s - 10ms/step - accuracy: 0.7264 - loss: 0.6645 -
val_accuracy: 0.8468 - val_loss: 0.6025 - learning_rate: 0.0100
Epoch 8/5000
11/11 - 0s - 9ms/step - accuracy: 0.8108 - loss: 0.5981 -
val_accuracy: 0.8401 - val_loss: 0.5568 - learning_rate: 0.0100
Epoch 9/5000
11/11 - 0s - 10ms/step - accuracy: 0.8491 - loss: 0.5415 -
val_accuracy: 0.8851 - val_loss: 0.4948 - learning_rate: 0.0100
Epoch 10/5000
11/11 - 0s - 9ms/step - accuracy: 0.8390 - loss: 0.4996 -
val_accuracy: 0.8806 - val_loss: 0.4584 - learning_rate: 0.0100
Epoch 11/5000
11/11 - 0s - 9ms/step - accuracy: 0.8649 - loss: 0.4686 -
val_accuracy: 0.8829 - val_loss: 0.4327 - learning_rate: 0.0100
Epoch 12/5000
11/11 - 0s - 9ms/step - accuracy: 0.8604 - loss: 0.4347 -
val_accuracy: 0.8874 - val_loss: 0.4034 - learning_rate: 0.0100
Epoch 13/5000
11/11 - 0s - 9ms/step - accuracy: 0.8699 - loss: 0.4083 -
val_accuracy: 0.8874 - val_loss: 0.3787 - learning_rate: 0.0100
Epoch 14/5000
11/11 - 0s - 9ms/step - accuracy: 0.8626 - loss: 0.3850 -
val_accuracy: 0.8919 - val_loss: 0.3582 - learning_rate: 0.0100
Epoch 15/5000
11/11 - 0s - 9ms/step - accuracy: 0.8784 - loss: 0.3649 -
val_accuracy: 0.8986 - val_loss: 0.3375 - learning_rate: 0.0100
Epoch 16/5000
11/11 - 0s - 9ms/step - accuracy: 0.8711 - loss: 0.3454 -
val_accuracy: 0.9054 - val_loss: 0.3194 - learning_rate: 0.0100
Epoch 17/5000
11/11 - 0s - 10ms/step - accuracy: 0.8992 - loss: 0.3302 -

val_accuracy: 0.9279 - val_loss: 0.3038 - learning_rate: 0.0100
Epoch 18/5000
11/11 - 0s - 9ms/step - accuracy: 0.9026 - loss: 0.3103 -
val_accuracy: 0.9189 - val_loss: 0.2892 - learning_rate: 0.0100
Epoch 19/5000
11/11 - 0s - 9ms/step - accuracy: 0.9184 - loss: 0.2950 -
val_accuracy: 0.9324 - val_loss: 0.2744 - learning_rate: 0.0100
Epoch 20/5000
11/11 - 0s - 9ms/step - accuracy: 0.9184 - loss: 0.2810 -
val_accuracy: 0.9279 - val_loss: 0.2603 - learning_rate: 0.0100
Epoch 21/5000
11/11 - 0s - 9ms/step - accuracy: 0.9167 - loss: 0.2703 -
val_accuracy: 0.9369 - val_loss: 0.2476 - learning_rate: 0.0100
Epoch 22/5000
11/11 - 0s - 9ms/step - accuracy: 0.9150 - loss: 0.2584 -
val_accuracy: 0.9347 - val_loss: 0.2380 - learning_rate: 0.0100
Epoch 23/5000
11/11 - 0s - 9ms/step - accuracy: 0.9257 - loss: 0.2451 -
val_accuracy: 0.9505 - val_loss: 0.2279 - learning_rate: 0.0100
Epoch 24/5000
11/11 - 0s - 9ms/step - accuracy: 0.9431 - loss: 0.2381 -
val_accuracy: 0.9459 - val_loss: 0.2213 - learning_rate: 0.0100
Epoch 25/5000
11/11 - 0s - 10ms/step - accuracy: 0.9189 - loss: 0.2310 -
val_accuracy: 0.9414 - val_loss: 0.2100 - learning_rate: 0.0100
Epoch 26/5000
11/11 - 0s - 9ms/step - accuracy: 0.9184 - loss: 0.2253 -
val_accuracy: 0.9482 - val_loss: 0.2031 - learning_rate: 0.0100
Epoch 27/5000
11/11 - 0s - 9ms/step - accuracy: 0.9516 - loss: 0.2107 -
val_accuracy: 0.9617 - val_loss: 0.1971 - learning_rate: 0.0100
Epoch 28/5000
11/11 - 0s - 9ms/step - accuracy: 0.9426 - loss: 0.2022 -
val_accuracy: 0.9459 - val_loss: 0.1894 - learning_rate: 0.0100
Epoch 29/5000
11/11 - 0s - 9ms/step - accuracy: 0.9341 - loss: 0.1974 -
val_accuracy: 0.9617 - val_loss: 0.1818 - learning_rate: 0.0100
Epoch 30/5000
11/11 - 0s - 9ms/step - accuracy: 0.9443 - loss: 0.1939 -
val_accuracy: 0.9595 - val_loss: 0.1753 - learning_rate: 0.0100
Epoch 31/5000
11/11 - 0s - 9ms/step - accuracy: 0.9347 - loss: 0.1845 -
val_accuracy: 0.9482 - val_loss: 0.1702 - learning_rate: 0.0100
Epoch 32/5000
11/11 - 0s - 9ms/step - accuracy: 0.9398 - loss: 0.1783 -
val_accuracy: 0.9775 - val_loss: 0.1660 - learning_rate: 0.0100
Epoch 33/5000
11/11 - 0s - 9ms/step - accuracy: 0.9651 - loss: 0.1740 -
val_accuracy: 0.9640 - val_loss: 0.1594 - learning_rate: 0.0100

Epoch 34/5000
11/11 - 0s - 9ms/step - accuracy: 0.9386 - loss: 0.1700 -
val_accuracy: 0.9640 - val_loss: 0.1550 - learning_rate: 0.0100
Epoch 35/5000
11/11 - 0s - 9ms/step - accuracy: 0.9657 - loss: 0.1632 -
val_accuracy: 0.9820 - val_loss: 0.1525 - learning_rate: 0.0100
Epoch 36/5000
11/11 - 0s - 9ms/step - accuracy: 0.9685 - loss: 0.1631 -
val_accuracy: 0.9617 - val_loss: 0.1510 - learning_rate: 0.0100
Epoch 37/5000
11/11 - 0s - 9ms/step - accuracy: 0.9352 - loss: 0.1598 -
val_accuracy: 0.9707 - val_loss: 0.1435 - learning_rate: 0.0100
Epoch 38/5000
11/11 - 0s - 10ms/step - accuracy: 0.9758 - loss: 0.1531 -
val_accuracy: 0.9797 - val_loss: 0.1419 - learning_rate: 0.0100
Epoch 39/5000
11/11 - 0s - 9ms/step - accuracy: 0.9668 - loss: 0.1489 -
val_accuracy: 0.9662 - val_loss: 0.1377 - learning_rate: 0.0100
Epoch 40/5000
11/11 - 0s - 9ms/step - accuracy: 0.9583 - loss: 0.1504 -
val_accuracy: 0.9707 - val_loss: 0.1412 - learning_rate: 0.0100
Epoch 41/5000
11/11 - 0s - 9ms/step - accuracy: 0.9685 - loss: 0.1434 -
val_accuracy: 0.9820 - val_loss: 0.1327 - learning_rate: 0.0100
Epoch 42/5000
11/11 - 0s - 9ms/step - accuracy: 0.9690 - loss: 0.1454 -
val_accuracy: 0.9730 - val_loss: 0.1305 - learning_rate: 0.0100
Epoch 43/5000
11/11 - 0s - 9ms/step - accuracy: 0.9482 - loss: 0.1398 -
val_accuracy: 0.9820 - val_loss: 0.1265 - learning_rate: 0.0100
Epoch 44/5000
11/11 - 0s - 9ms/step - accuracy: 0.9769 - loss: 0.1364 -
val_accuracy: 0.9797 - val_loss: 0.1260 - learning_rate: 0.0100
Epoch 45/5000
11/11 - 0s - 10ms/step - accuracy: 0.9443 - loss: 0.1363 -
val_accuracy: 0.9752 - val_loss: 0.1217 - learning_rate: 0.0100
Epoch 46/5000
11/11 - 0s - 9ms/step - accuracy: 0.9718 - loss: 0.1317 -
val_accuracy: 0.9842 - val_loss: 0.1205 - learning_rate: 0.0100
Epoch 47/5000
11/11 - 0s - 9ms/step - accuracy: 0.9741 - loss: 0.1272 -
val_accuracy: 0.9775 - val_loss: 0.1168 - learning_rate: 0.0100
Epoch 48/5000
11/11 - 0s - 10ms/step - accuracy: 0.9645 - loss: 0.1262 -
val_accuracy: 0.9842 - val_loss: 0.1190 - learning_rate: 0.0100
Epoch 49/5000
11/11 - 0s - 9ms/step - accuracy: 0.9690 - loss: 0.1273 -
val_accuracy: 0.9797 - val_loss: 0.1146 - learning_rate: 0.0100
Epoch 50/5000

11/11 - 0s - 9ms/step - accuracy: 0.9623 - loss: 0.1231 -
val_accuracy: 0.9752 - val_loss: 0.1121 - learning_rate: 0.0100
Epoch 51/5000
11/11 - 0s - 9ms/step - accuracy: 0.9758 - loss: 0.1188 -
val_accuracy: 0.9842 - val_loss: 0.1105 - learning_rate: 0.0100
Epoch 52/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.1163 -
val_accuracy: 0.9797 - val_loss: 0.1076 - learning_rate: 0.0100
Epoch 53/5000
11/11 - 0s - 9ms/step - accuracy: 0.9679 - loss: 0.1180 -
val_accuracy: 0.9842 - val_loss: 0.1070 - learning_rate: 0.0100
Epoch 54/5000
11/11 - 0s - 8ms/step - accuracy: 0.9769 - loss: 0.1143 -
val_accuracy: 0.9797 - val_loss: 0.1051 - learning_rate: 0.0100
Epoch 55/5000
11/11 - 0s - 9ms/step - accuracy: 0.9730 - loss: 0.1126 -
val_accuracy: 0.9775 - val_loss: 0.1041 - learning_rate: 0.0100
Epoch 56/5000
11/11 - 0s - 9ms/step - accuracy: 0.9758 - loss: 0.1117 -
val_accuracy: 0.9797 - val_loss: 0.1048 - learning_rate: 0.0100
Epoch 57/5000
11/11 - 0s - 10ms/step - accuracy: 0.9741 - loss: 0.1103 -
val_accuracy: 0.9820 - val_loss: 0.1000 - learning_rate: 0.0100
Epoch 58/5000
11/11 - 0s - 9ms/step - accuracy: 0.9657 - loss: 0.1104 -
val_accuracy: 0.9842 - val_loss: 0.0995 - learning_rate: 0.0100
Epoch 59/5000
11/11 - 0s - 9ms/step - accuracy: 0.9589 - loss: 0.1142 -
val_accuracy: 0.9820 - val_loss: 0.1043 - learning_rate: 0.0100
Epoch 60/5000
11/11 - 0s - 9ms/step - accuracy: 0.9516 - loss: 0.1135 -
val_accuracy: 0.9752 - val_loss: 0.0969 - learning_rate: 0.0100
Epoch 61/5000
11/11 - 0s - 9ms/step - accuracy: 0.9718 - loss: 0.1072 -
val_accuracy: 0.9707 - val_loss: 0.0994 - learning_rate: 0.0100
Epoch 62/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.1041 -
val_accuracy: 0.9752 - val_loss: 0.0960 - learning_rate: 0.0100
Epoch 63/5000
11/11 - 0s - 9ms/step - accuracy: 0.9702 - loss: 0.1045 -
val_accuracy: 0.9842 - val_loss: 0.0944 - learning_rate: 0.0100
Epoch 64/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.1010 -
val_accuracy: 0.9820 - val_loss: 0.0926 - learning_rate: 0.0100
Epoch 65/5000
11/11 - 0s - 9ms/step - accuracy: 0.9566 - loss: 0.1051 -
val_accuracy: 0.9820 - val_loss: 0.0934 - learning_rate: 0.0100
Epoch 66/5000
11/11 - 0s - 10ms/step - accuracy: 0.9628 - loss: 0.1038 -

val_accuracy: 0.9775 - val_loss: 0.0938 - learning_rate: 0.0100
Epoch 67/5000
11/11 - 0s - 9ms/step - accuracy: 0.9730 - loss: 0.0998 -
val_accuracy: 0.9842 - val_loss: 0.0908 - learning_rate: 0.0100
Epoch 68/5000
11/11 - 0s - 9ms/step - accuracy: 0.9769 - loss: 0.0985 -
val_accuracy: 0.9842 - val_loss: 0.0901 - learning_rate: 0.0100
Epoch 69/5000
11/11 - 0s - 9ms/step - accuracy: 0.9713 - loss: 0.0982 -
val_accuracy: 0.9820 - val_loss: 0.0883 - learning_rate: 0.0100
Epoch 70/5000
11/11 - 0s - 9ms/step - accuracy: 0.9668 - loss: 0.1013 -
val_accuracy: 0.9820 - val_loss: 0.0890 - learning_rate: 0.0100
Epoch 71/5000
11/11 - 0s - 9ms/step - accuracy: 0.9662 - loss: 0.0990 -
val_accuracy: 0.9797 - val_loss: 0.0867 - learning_rate: 0.0100
Epoch 72/5000
11/11 - 0s - 9ms/step - accuracy: 0.9764 - loss: 0.0939 -
val_accuracy: 0.9775 - val_loss: 0.0862 - learning_rate: 0.0100
Epoch 73/5000
11/11 - 0s - 9ms/step - accuracy: 0.9758 - loss: 0.0938 -
val_accuracy: 0.9775 - val_loss: 0.0880 - learning_rate: 0.0100
Epoch 74/5000
11/11 - 0s - 9ms/step - accuracy: 0.9769 - loss: 0.0922 -
val_accuracy: 0.9820 - val_loss: 0.0859 - learning_rate: 0.0100
Epoch 75/5000
11/11 - 0s - 9ms/step - accuracy: 0.9769 - loss: 0.0921 -
val_accuracy: 0.9842 - val_loss: 0.0851 - learning_rate: 0.0100
Epoch 76/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.0998 -
val_accuracy: 0.9820 - val_loss: 0.0891 - learning_rate: 0.0100
Epoch 77/5000
11/11 - 0s - 9ms/step - accuracy: 0.9792 - loss: 0.0917 -
val_accuracy: 0.9797 - val_loss: 0.0857 - learning_rate: 0.0100
Epoch 78/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0894 -
val_accuracy: 0.9820 - val_loss: 0.0843 - learning_rate: 0.0100
Epoch 79/5000
11/11 - 0s - 9ms/step - accuracy: 0.9780 - loss: 0.0883 -
val_accuracy: 0.9820 - val_loss: 0.0834 - learning_rate: 0.0100
Epoch 80/5000
11/11 - 0s - 11ms/step - accuracy: 0.9758 - loss: 0.0888 -
val_accuracy: 0.9842 - val_loss: 0.0808 - learning_rate: 0.0100
Epoch 81/5000
11/11 - 0s - 9ms/step - accuracy: 0.9730 - loss: 0.0930 -
val_accuracy: 0.9775 - val_loss: 0.0799 - learning_rate: 0.0100
Epoch 82/5000
11/11 - 0s - 9ms/step - accuracy: 0.9809 - loss: 0.0859 -
val_accuracy: 0.9775 - val_loss: 0.0857 - learning_rate: 0.0100

Epoch 83/5000
11/11 - 0s - 9ms/step - accuracy: 0.9690 - loss: 0.0890 -
val_accuracy: 0.9842 - val_loss: 0.0789 - learning_rate: 0.0100
Epoch 84/5000
11/11 - 0s - 9ms/step - accuracy: 0.9702 - loss: 0.0897 -
val_accuracy: 0.9797 - val_loss: 0.0801 - learning_rate: 0.0100
Epoch 85/5000
11/11 - 0s - 10ms/step - accuracy: 0.9741 - loss: 0.0858 -
val_accuracy: 0.9797 - val_loss: 0.0803 - learning_rate: 0.0100
Epoch 86/5000
11/11 - 0s - 9ms/step - accuracy: 0.9780 - loss: 0.0856 -
val_accuracy: 0.9820 - val_loss: 0.0785 - learning_rate: 0.0100
Epoch 87/5000
11/11 - 0s - 9ms/step - accuracy: 0.9657 - loss: 0.0870 -
val_accuracy: 0.9775 - val_loss: 0.0811 - learning_rate: 0.0100
Epoch 88/5000
11/11 - 0s - 10ms/step - accuracy: 0.9747 - loss: 0.0836 -
val_accuracy: 0.9797 - val_loss: 0.0778 - learning_rate: 0.0100
Epoch 89/5000
11/11 - 0s - 9ms/step - accuracy: 0.9713 - loss: 0.0846 -
val_accuracy: 0.9797 - val_loss: 0.0784 - learning_rate: 0.0100
Epoch 90/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0814 -
val_accuracy: 0.9775 - val_loss: 0.0782 - learning_rate: 0.0100
Epoch 91/5000
11/11 - 0s - 9ms/step - accuracy: 0.9724 - loss: 0.0829 -
val_accuracy: 0.9820 - val_loss: 0.0763 - learning_rate: 0.0100
Epoch 92/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0815 -
val_accuracy: 0.9842 - val_loss: 0.0764 - learning_rate: 0.0100
Epoch 93/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0804 -
val_accuracy: 0.9775 - val_loss: 0.0772 - learning_rate: 0.0100
Epoch 94/5000
11/11 - 0s - 9ms/step - accuracy: 0.9747 - loss: 0.0815 -
val_accuracy: 0.9865 - val_loss: 0.0738 - learning_rate: 0.0100
Epoch 95/5000
11/11 - 0s - 9ms/step - accuracy: 0.9730 - loss: 0.0834 -
val_accuracy: 0.9842 - val_loss: 0.0741 - learning_rate: 0.0100
Epoch 96/5000
11/11 - 0s - 10ms/step - accuracy: 0.9803 - loss: 0.0784 -
val_accuracy: 0.9775 - val_loss: 0.0736 - learning_rate: 0.0100
Epoch 97/5000
11/11 - 0s - 14ms/step - accuracy: 0.9814 - loss: 0.0780 -
val_accuracy: 0.9820 - val_loss: 0.0763 - learning_rate: 0.0100
Epoch 98/5000
11/11 - 0s - 10ms/step - accuracy: 0.9758 - loss: 0.0800 -
val_accuracy: 0.9820 - val_loss: 0.0746 - learning_rate: 0.0100
Epoch 99/5000

11/11 - 0s - 11ms/step - accuracy: 0.9803 - loss: 0.0773 -
val_accuracy: 0.9775 - val_loss: 0.0720 - learning_rate: 0.0100
Epoch 100/5000
11/11 - 0s - 10ms/step - accuracy: 0.9803 - loss: 0.0776 -
val_accuracy: 0.9797 - val_loss: 0.0730 - learning_rate: 0.0100
Epoch 101/5000
11/11 - 0s - 9ms/step - accuracy: 0.9764 - loss: 0.0776 -
val_accuracy: 0.9797 - val_loss: 0.0721 - learning_rate: 0.0100
Epoch 102/5000
11/11 - 0s - 9ms/step - accuracy: 0.9809 - loss: 0.0760 -
val_accuracy: 0.9820 - val_loss: 0.0695 - learning_rate: 0.0100
Epoch 103/5000
11/11 - 0s - 12ms/step - accuracy: 0.9662 - loss: 0.0940 -
val_accuracy: 0.9775 - val_loss: 0.0706 - learning_rate: 0.0100
Epoch 104/5000
11/11 - 0s - 9ms/step - accuracy: 0.9758 - loss: 0.0777 -
val_accuracy: 0.9730 - val_loss: 0.0845 - learning_rate: 0.0100
Epoch 105/5000
11/11 - 0s - 8ms/step - accuracy: 0.9764 - loss: 0.0779 -
val_accuracy: 0.9820 - val_loss: 0.0689 - learning_rate: 0.0100
Epoch 106/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0762 -
val_accuracy: 0.9797 - val_loss: 0.0707 - learning_rate: 0.0100
Epoch 107/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0739 -
val_accuracy: 0.9820 - val_loss: 0.0698 - learning_rate: 0.0100
Epoch 108/5000
11/11 - 0s - 9ms/step - accuracy: 0.9814 - loss: 0.0729 -
val_accuracy: 0.9775 - val_loss: 0.0743 - learning_rate: 0.0100
Epoch 109/5000
11/11 - 0s - 8ms/step - accuracy: 0.9679 - loss: 0.0825 -
val_accuracy: 0.9797 - val_loss: 0.0732 - learning_rate: 0.0100
Epoch 110/5000
11/11 - 0s - 9ms/step - accuracy: 0.9679 - loss: 0.0802 -
val_accuracy: 0.9842 - val_loss: 0.0689 - learning_rate: 0.0100
Epoch 111/5000
11/11 - 0s - 8ms/step - accuracy: 0.9640 - loss: 0.0782 -
val_accuracy: 0.9775 - val_loss: 0.0739 - learning_rate: 0.0100
Epoch 112/5000
11/11 - 0s - 9ms/step - accuracy: 0.9792 - loss: 0.0721 -
val_accuracy: 0.9842 - val_loss: 0.0673 - learning_rate: 0.0100
Epoch 113/5000
11/11 - 0s - 10ms/step - accuracy: 0.9758 - loss: 0.0722 -
val_accuracy: 0.9797 - val_loss: 0.0689 - learning_rate: 0.0100
Epoch 114/5000
11/11 - 0s - 9ms/step - accuracy: 0.9803 - loss: 0.0713 -
val_accuracy: 0.9775 - val_loss: 0.0720 - learning_rate: 0.0100
Epoch 115/5000
11/11 - 0s - 9ms/step - accuracy: 0.9566 - loss: 0.0843 -

val_accuracy: 0.9797 - val_loss: 0.0723 - learning_rate: 0.0100
Epoch 116/5000
11/11 - 0s - 9ms/step - accuracy: 0.9747 - loss: 0.0741 -
val_accuracy: 0.9797 - val_loss: 0.0674 - learning_rate: 0.0100
Epoch 117/5000
11/11 - 0s - 8ms/step - accuracy: 0.9640 - loss: 0.0789 -
val_accuracy: 0.9797 - val_loss: 0.0676 - learning_rate: 0.0100
Epoch 118/5000
11/11 - 0s - 9ms/step - accuracy: 0.9758 - loss: 0.0764 -
val_accuracy: 0.9820 - val_loss: 0.0673 - learning_rate: 0.0100
Epoch 119/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.0699 -
val_accuracy: 0.9797 - val_loss: 0.0690 - learning_rate: 0.0100
Epoch 120/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0706 -
val_accuracy: 0.9842 - val_loss: 0.0658 - learning_rate: 0.0100
Epoch 121/5000
11/11 - 0s - 9ms/step - accuracy: 0.9809 - loss: 0.0693 -
val_accuracy: 0.9820 - val_loss: 0.0666 - learning_rate: 0.0100
Epoch 122/5000
11/11 - 0s - 10ms/step - accuracy: 0.9825 - loss: 0.0685 -
val_accuracy: 0.9775 - val_loss: 0.0695 - learning_rate: 0.0100
Epoch 123/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0695 -
val_accuracy: 0.9820 - val_loss: 0.0676 - learning_rate: 0.0100
Epoch 124/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0689 -
val_accuracy: 0.9842 - val_loss: 0.0634 - learning_rate: 0.0100
Epoch 125/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0709 -
val_accuracy: 0.9820 - val_loss: 0.0675 - learning_rate: 0.0100
Epoch 126/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0694 -
val_accuracy: 0.9797 - val_loss: 0.0675 - learning_rate: 0.0100
Epoch 127/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.0693 -
val_accuracy: 0.9797 - val_loss: 0.0653 - learning_rate: 0.0100
Epoch 128/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.0690 -
val_accuracy: 0.9797 - val_loss: 0.0671 - learning_rate: 0.0100
Epoch 129/5000
11/11 - 0s - 9ms/step - accuracy: 0.9814 - loss: 0.0667 -
val_accuracy: 0.9865 - val_loss: 0.0629 - learning_rate: 0.0100
Epoch 130/5000
11/11 - 0s - 8ms/step - accuracy: 0.9797 - loss: 0.0667 -
val_accuracy: 0.9797 - val_loss: 0.0655 - learning_rate: 0.0100
Epoch 131/5000
11/11 - 0s - 8ms/step - accuracy: 0.9803 - loss: 0.0665 -
val_accuracy: 0.9820 - val_loss: 0.0652 - learning_rate: 0.0100

Epoch 132/5000
11/11 - 0s - 8ms/step - accuracy: 0.9775 - loss: 0.0677 -
val_accuracy: 0.9820 - val_loss: 0.0622 - learning_rate: 0.0100
Epoch 133/5000
11/11 - 0s - 8ms/step - accuracy: 0.9809 - loss: 0.0655 -
val_accuracy: 0.9820 - val_loss: 0.0639 - learning_rate: 0.0100
Epoch 134/5000
11/11 - 0s - 8ms/step - accuracy: 0.9831 - loss: 0.0651 -
val_accuracy: 0.9820 - val_loss: 0.0628 - learning_rate: 0.0100
Epoch 135/5000
11/11 - 0s - 8ms/step - accuracy: 0.9814 - loss: 0.0653 -
val_accuracy: 0.9797 - val_loss: 0.0605 - learning_rate: 0.0100
Epoch 136/5000
11/11 - 0s - 9ms/step - accuracy: 0.9747 - loss: 0.0694 -
val_accuracy: 0.9797 - val_loss: 0.0606 - learning_rate: 0.0100
Epoch 137/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0657 -
val_accuracy: 0.9820 - val_loss: 0.0673 - learning_rate: 0.0100
Epoch 138/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0646 -
val_accuracy: 0.9797 - val_loss: 0.0631 - learning_rate: 0.0100
Epoch 139/5000
11/11 - 0s - 9ms/step - accuracy: 0.9769 - loss: 0.0664 -
val_accuracy: 0.9842 - val_loss: 0.0610 - learning_rate: 0.0100
Epoch 140/5000
11/11 - 0s - 9ms/step - accuracy: 0.9775 - loss: 0.0676 -
val_accuracy: 0.9775 - val_loss: 0.0639 - learning_rate: 0.0100
Epoch 141/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0647 -
val_accuracy: 0.9797 - val_loss: 0.0609 - learning_rate: 0.0100
Epoch 142/5000
11/11 - 0s - 9ms/step - accuracy: 0.9803 - loss: 0.0656 -
val_accuracy: 0.9842 - val_loss: 0.0618 - learning_rate: 0.0100
Epoch 143/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0654 -
val_accuracy: 0.9820 - val_loss: 0.0636 - learning_rate: 0.0100
Epoch 144/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0646 -
val_accuracy: 0.9797 - val_loss: 0.0656 - learning_rate: 0.0100
Epoch 145/5000
11/11 - 0s - 9ms/step - accuracy: 0.9792 - loss: 0.0656 -
val_accuracy: 0.9775 - val_loss: 0.0645 - learning_rate: 1.0000e-03
Epoch 146/5000
11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0643 -
val_accuracy: 0.9842 - val_loss: 0.0621 - learning_rate: 1.0000e-03
Epoch 147/5000
11/11 - 0s - 9ms/step - accuracy: 0.9786 - loss: 0.0635 -
val_accuracy: 0.9865 - val_loss: 0.0611 - learning_rate: 1.0000e-03
Epoch 148/5000

11/11 - 0s - 9ms/step - accuracy: 0.9797 - loss: 0.0626 -
val_accuracy: 0.9820 - val_loss: 0.0606 - learning_rate: 1.0000e-03
Epoch 149/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0626 -
val_accuracy: 0.9820 - val_loss: 0.0597 - learning_rate: 1.0000e-03
Epoch 150/5000
11/11 - 0s - 11ms/step - accuracy: 0.9831 - loss: 0.0622 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-03
Epoch 151/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0622 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-03
Epoch 152/5000
11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0621 -
val_accuracy: 0.9820 - val_loss: 0.0608 - learning_rate: 1.0000e-03
Epoch 153/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0622 -
val_accuracy: 0.9797 - val_loss: 0.0612 - learning_rate: 1.0000e-03
Epoch 154/5000
11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0621 -
val_accuracy: 0.9797 - val_loss: 0.0610 - learning_rate: 1.0000e-03
Epoch 155/5000
11/11 - 0s - 8ms/step - accuracy: 0.9837 - loss: 0.0623 -
val_accuracy: 0.9797 - val_loss: 0.0613 - learning_rate: 1.0000e-03
Epoch 156/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0620 -
val_accuracy: 0.9797 - val_loss: 0.0616 - learning_rate: 1.0000e-03
Epoch 157/5000
11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0619 -
val_accuracy: 0.9820 - val_loss: 0.0615 - learning_rate: 1.0000e-03
Epoch 158/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0619 -
val_accuracy: 0.9797 - val_loss: 0.0616 - learning_rate: 1.0000e-03
Epoch 159/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0619 -
val_accuracy: 0.9797 - val_loss: 0.0613 - learning_rate: 1.0000e-03
Epoch 160/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0619 -
val_accuracy: 0.9797 - val_loss: 0.0612 - learning_rate: 1.0000e-03
Epoch 161/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0620 -
val_accuracy: 0.9820 - val_loss: 0.0613 - learning_rate: 1.0000e-03
Epoch 162/5000
11/11 - 0s - 8ms/step - accuracy: 0.9814 - loss: 0.0622 -
val_accuracy: 0.9775 - val_loss: 0.0639 - learning_rate: 1.0000e-03
Epoch 163/5000
11/11 - 0s - 8ms/step - accuracy: 0.9825 - loss: 0.0624 -
val_accuracy: 0.9775 - val_loss: 0.0641 - learning_rate: 1.0000e-03
Epoch 164/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0622 -
val_accuracy: 0.9775 - val_loss: 0.0629 - learning_rate: 1.0000e-03

Epoch 165/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0620 -
val_accuracy: 0.9797 - val_loss: 0.0599 - learning_rate: 1.0000e-03
Epoch 166/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0621 -
val_accuracy: 0.9797 - val_loss: 0.0597 - learning_rate: 1.0000e-03
Epoch 167/5000
11/11 - 0s - 9ms/step - accuracy: 0.9814 - loss: 0.0617 -
val_accuracy: 0.9797 - val_loss: 0.0619 - learning_rate: 1.0000e-03
Epoch 168/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0619 -
val_accuracy: 0.9797 - val_loss: 0.0599 - learning_rate: 1.0000e-03
Epoch 169/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0619 -
val_accuracy: 0.9797 - val_loss: 0.0594 - learning_rate: 1.0000e-03
Epoch 170/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0619 -
val_accuracy: 0.9820 - val_loss: 0.0593 - learning_rate: 1.0000e-03
Epoch 171/5000
11/11 - 0s - 9ms/step - accuracy: 0.9814 - loss: 0.0619 -
val_accuracy: 0.9842 - val_loss: 0.0595 - learning_rate: 1.0000e-03
Epoch 172/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0619 -
val_accuracy: 0.9842 - val_loss: 0.0598 - learning_rate: 1.0000e-03
Epoch 173/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0618 -
val_accuracy: 0.9820 - val_loss: 0.0605 - learning_rate: 1.0000e-03
Epoch 174/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0616 -
val_accuracy: 0.9842 - val_loss: 0.0607 - learning_rate: 1.0000e-03
Epoch 175/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0616 -
val_accuracy: 0.9842 - val_loss: 0.0609 - learning_rate: 1.0000e-03
Epoch 176/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0616 -
val_accuracy: 0.9842 - val_loss: 0.0608 - learning_rate: 1.0000e-03
Epoch 177/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0619 -
val_accuracy: 0.9842 - val_loss: 0.0602 - learning_rate: 1.0000e-03
Epoch 178/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0617 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-03
Epoch 179/5000
11/11 - 0s - 9ms/step - accuracy: 0.9825 - loss: 0.0615 -
val_accuracy: 0.9820 - val_loss: 0.0606 - learning_rate: 1.0000e-03
Epoch 180/5000
11/11 - 0s - 10ms/step - accuracy: 0.9837 - loss: 0.0614 -
val_accuracy: 0.9820 - val_loss: 0.0606 - learning_rate: 1.0000e-03
Epoch 181/5000

11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0614 -
val_accuracy: 0.9797 - val_loss: 0.0610 - learning_rate: 1.0000e-03
Epoch 182/5000
11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0615 -
val_accuracy: 0.9842 - val_loss: 0.0599 - learning_rate: 1.0000e-03
Epoch 183/5000
11/11 - 0s - 8ms/step - accuracy: 0.9820 - loss: 0.0614 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-03
Epoch 184/5000
11/11 - 0s - 8ms/step - accuracy: 0.9820 - loss: 0.0614 -
val_accuracy: 0.9842 - val_loss: 0.0610 - learning_rate: 1.0000e-03
Epoch 185/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0619 -
val_accuracy: 0.9775 - val_loss: 0.0634 - learning_rate: 1.0000e-03
Epoch 186/5000
11/11 - 0s - 9ms/step - accuracy: 0.9820 - loss: 0.0618 -
val_accuracy: 0.9797 - val_loss: 0.0632 - learning_rate: 1.0000e-03
Epoch 187/5000
11/11 - 0s - 9ms/step - accuracy: 0.9814 - loss: 0.0616 -
val_accuracy: 0.9797 - val_loss: 0.0621 - learning_rate: 1.0000e-03
Epoch 188/5000
11/11 - 0s - 8ms/step - accuracy: 0.9825 - loss: 0.0614 -
val_accuracy: 0.9820 - val_loss: 0.0612 - learning_rate: 1.0000e-03
Epoch 189/5000
11/11 - 0s - 8ms/step - accuracy: 0.9820 - loss: 0.0611 -
val_accuracy: 0.9842 - val_loss: 0.0603 - learning_rate: 1.0000e-03
Epoch 190/5000
11/11 - 0s - 8ms/step - accuracy: 0.9814 - loss: 0.0614 -
val_accuracy: 0.9842 - val_loss: 0.0603 - learning_rate: 1.0000e-03
Epoch 191/5000
11/11 - 0s - 8ms/step - accuracy: 0.9825 - loss: 0.0614 -
val_accuracy: 0.9865 - val_loss: 0.0598 - learning_rate: 1.0000e-03
Epoch 192/5000
11/11 - 0s - 10ms/step - accuracy: 0.9814 - loss: 0.0614 -
val_accuracy: 0.9865 - val_loss: 0.0589 - learning_rate: 1.0000e-03
Epoch 193/5000
11/11 - 0s - 8ms/step - accuracy: 0.9809 - loss: 0.0612 -
val_accuracy: 0.9820 - val_loss: 0.0595 - learning_rate: 1.0000e-03
Epoch 194/5000
11/11 - 0s - 9ms/step - accuracy: 0.9831 - loss: 0.0611 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-03
Epoch 195/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0605 - learning_rate: 1.0000e-04
Epoch 196/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0605 - learning_rate: 1.0000e-04
Epoch 197/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -

```
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 198/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 199/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 200/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 201/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 202/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 203/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 204/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 205/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 206/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 207/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 208/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 209/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 210/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 211/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0605 - learning_rate: 1.0000e-04
Epoch 212/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0605 - learning_rate: 1.0000e-04
Epoch 213/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9797 - val_loss: 0.0604 - learning_rate: 1.0000e-04
```

Epoch 214/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 215/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 216/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 217/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 218/5000
11/11 - 0s - 10ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 219/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-04
Epoch 220/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 221/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 222/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 223/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 224/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 225/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0599 - learning_rate: 1.0000e-04
Epoch 226/5000
11/11 - 0s - 10ms/step - accuracy: 0.9837 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0598 - learning_rate: 1.0000e-04
Epoch 227/5000
11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0610 -
val_accuracy: 0.9820 - val_loss: 0.0599 - learning_rate: 1.0000e-04
Epoch 228/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0599 - learning_rate: 1.0000e-04
Epoch 229/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0599 - learning_rate: 1.0000e-04
Epoch 230/5000

11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-04
Epoch 231/5000

11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-04
Epoch 232/5000

11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-04
Epoch 233/5000

11/11 - 0s - 10ms/step - accuracy: 0.9837 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-04
Epoch 234/5000

11/11 - 0s - 9ms/step - accuracy: 0.9837 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0600 - learning_rate: 1.0000e-04
Epoch 235/5000

11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0601 - learning_rate: 1.0000e-04
Epoch 236/5000

11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0601 - learning_rate: 1.0000e-04
Epoch 237/5000

11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0601 - learning_rate: 1.0000e-04
Epoch 238/5000

11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 239/5000

11/11 - 0s - 16ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0601 - learning_rate: 1.0000e-04
Epoch 240/5000

11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9797 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 241/5000

11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 242/5000

11/11 - 0s - 11ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0602 - learning_rate: 1.0000e-04
Epoch 243/5000

11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 244/5000

11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-04
Epoch 245/5000

11/11 - 0s - 10ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 246/5000

11/11 - 0s - 10ms/step - accuracy: 0.9848 - loss: 0.0609 -

[illegible]

Epoch 263/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 264/5000
11/11 - 0s - 9ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 265/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 266/5000
11/11 - 0s - 8ms/step - accuracy: 0.9848 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 267/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 268/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 269/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 270/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 271/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0603 - learning_rate: 1.0000e-05
Epoch 272/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 273/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 274/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 275/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 276/5000
11/11 - 0s - 9ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 277/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05
Epoch 278/5000
11/11 - 0s - 8ms/step - accuracy: 0.9842 - loss: 0.0609 -
val_accuracy: 0.9820 - val_loss: 0.0604 - learning_rate: 1.0000e-05

Training time : 0:00:28.843712
End of training

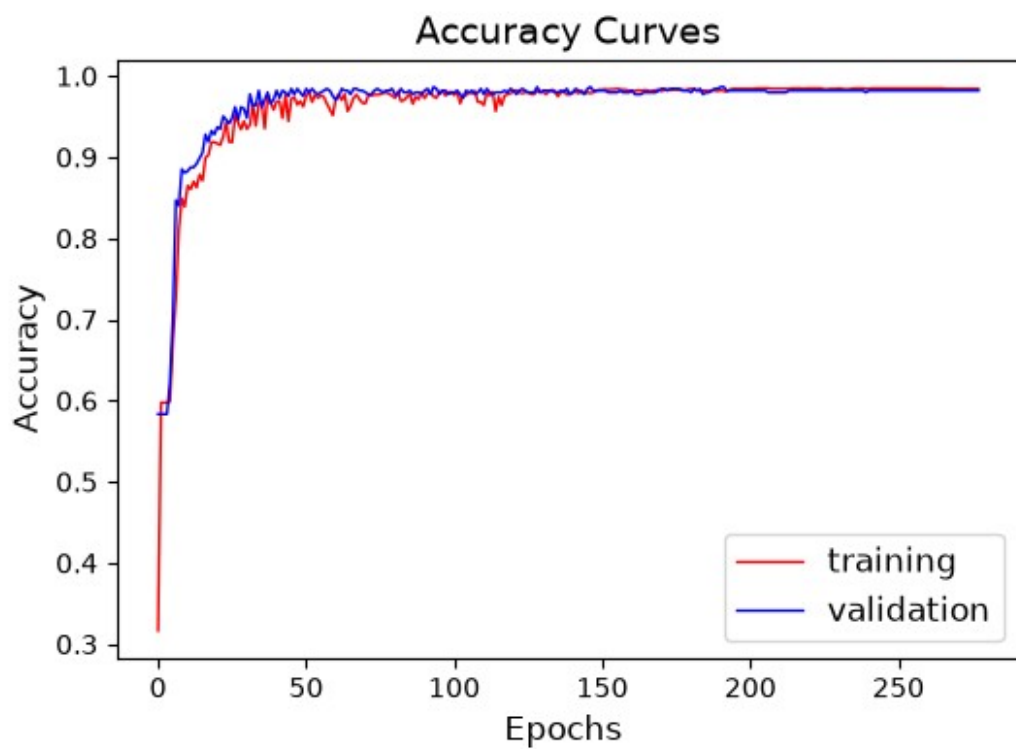
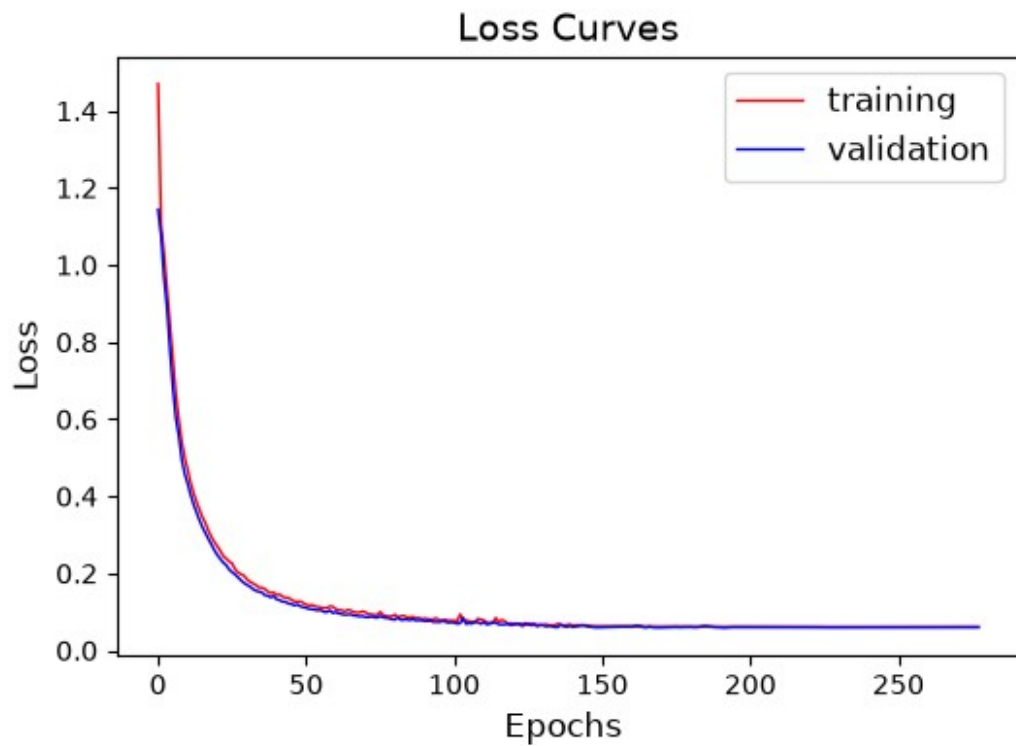
(5) Performance Evaluation

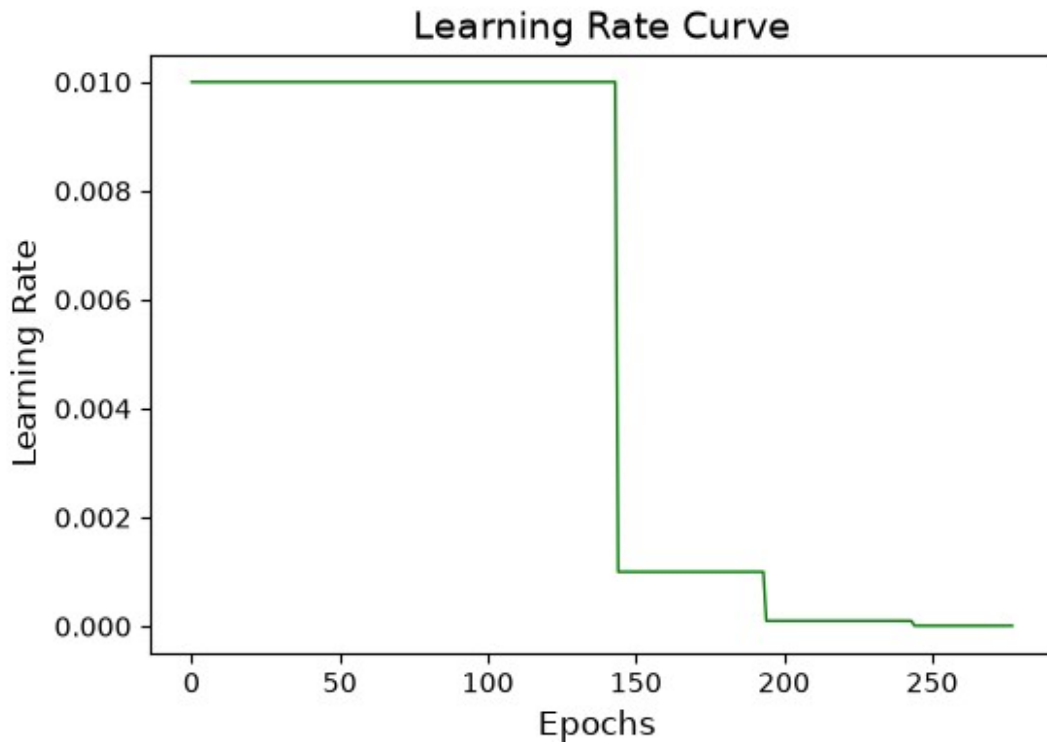
Training

```
plt.figure(figsize=[6, 4])
plt.plot(model_history.history['loss'], 'r', linewidth=1.0)
plt.plot(model_history.history['val_loss'], 'b', linewidth=1.0)
plt.legend(['training', 'validation'], fontsize=12)
plt.xlabel('Epochs ', fontsize=12)
plt.ylabel('Loss', fontsize=12)
plt.title('Loss Curves', fontsize=14)
plt.show()

plt.figure(figsize=[6, 4])
plt.plot(model_history.history['accuracy'], 'r', linewidth=1.0)
plt.plot(model_history.history['val_accuracy'], 'b', linewidth=1.0)
plt.legend(['training', 'validation'], fontsize=12)
plt.xlabel('Epochs ', fontsize=12)
plt.ylabel('Accuracy', fontsize=12)
plt.title('Accuracy Curves', fontsize=14)
plt.show()

plt.figure(figsize=[6, 4])
plt.plot(model_history.history['learning_rate'], 'g', linewidth=1.0)
plt.xlabel('Epochs ', fontsize=12)
plt.ylabel('Learning Rate', fontsize=12)
plt.title('Learning Rate Curve', fontsize=14)
plt.show()
```





Confusion Matrix

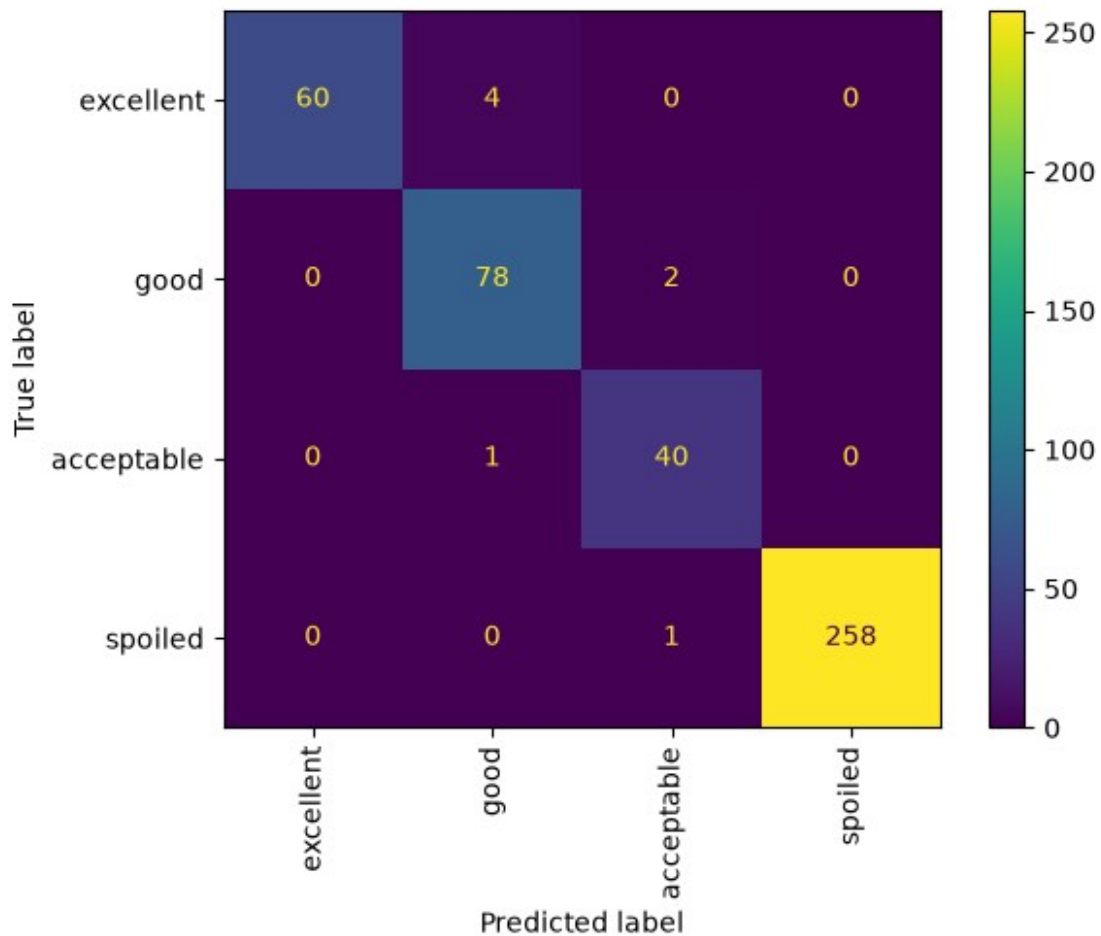
```
from sklearn import metrics

output_label = ['excellent', 'good', 'acceptable', 'spoiled']
pred_test_encode = model.predict(X_test_tensor)

pred_test = np.argmax(pred_test_encode, axis=1)
out_test = np.argmax(Y_test, axis=1)

cm_test = metrics.confusion_matrix(out_test, pred_test)
cm_test_display = metrics.ConfusionMatrixDisplay(
    confusion_matrix = cm_test, display_labels = output_label)
cm_test_display.plot(xticks_rotation='vertical')
plt.show()
```

14/14 ————— 0s 3ms/step



Precision, Recall, F1 Score

```
print(metrics.classification_report(out_test, pred_test,
target_names=output_label))
```

	precision	recall	f1-score	support
excellent	1.00	0.94	0.97	64
good	0.94	0.97	0.96	80
acceptable	0.93	0.98	0.95	41
spoiled	1.00	1.00	1.00	259
accuracy			0.98	444
macro avg	0.97	0.97	0.97	444
weighted avg	0.98	0.98	0.98	444

(6) Explore the Model

Model weights and biases

```
dense_layer1 = model.layers[0]
weights1, biases1 = dense_layer1.get_weights()
print(f'Weights_1 shape: {weights1.shape}')
print(f'Biases_1 shape: {biases1.shape}')
print(f'Weights_1:\n {weights1}')
print(f'Biases_1 :\n {biases1}')
```

```
dense_layer2 = model.layers[1]
weights2, biases2 = dense_layer2.get_weights()
print(f'Weights_2 shape: {weights2.shape}')
print(f'Biases_2 shape: {biases2.shape}')
print(f'Weights_2:\n {weights2}')
print(f'Biases_2 :\n {biases2}')
```

Weights_1 shape: (11, 20)

Biases_1 shape: (20,)

Weights_1:

```
[[-0.49673086 -0.19195022 -0.32001507 -0.02858155  0.43984812
 0.7789703
 0.10870647  0.20640269 -0.50477105 -0.09112339 -0.10905071 -
0.27004254
-0.5669689 -0.5384777  0.5710091  -0.5055324  -0.07681112 -
0.75858945
-1.2053697  0.5184446 ]
[-2.159873  0.9675849 -0.17105617  0.7557967  1.9412398
2.6548524
 3.2615018  1.8746587 -2.7193701 -1.3836765  2.583726 -
3.0332994
-4.7570004 -3.4170406  1.5769895 -1.6546324 -1.9762548 -
4.0742784
-2.2264628  1.2199585 ]
[ 2.5584323 -0.20999661  1.0494083 -3.3146284 -3.9575155 -
1.1941131
-2.7277877 -4.0410614  2.0802305  2.7255602 -3.4347076
3.1309986
 1.1041434  3.7084694 -5.0869036  4.0414963  4.377557
0.37169775
 1.4106237 -3.5813832 ]
[ 0.04895907  1.1626585  0.25084272  0.40738976  0.36152768
1.276067
 0.4637997  0.11617278 -0.42619213 -0.2756778 -0.04402969 -
0.26059878
-1.8352346  0.19262609  0.40768546 -0.09978434 -0.30073425 -
1.6811395
-1.6450247  0.4511208 ]
[ 0.4491912  0.30394092 -0.12565427  0.18505536  0.25531667
0.8337443
 0.32153434 -0.01880024  0.22328328  0.02916905  0.28408352 -
0.35123867
-0.85260326 -0.4472914  0.08844823 -0.24058542 -0.03358822 -
```

```

0.1702035
-0.42877403 0.31955427]
[ 0.29142317 1.2682575 0.2643152 -0.18175797 -0.17912841
1.7147954
-0.05853544 0.3530866 0.5723523 0.3509084 -0.19225045 -
0.16310515
-1.6874785 0.53740406 0.3920495 -0.39773622 -0.3361777 -
0.88416207
-1.0848702 0.06507642]
[-1.4486791 1.132816 0.31791213 1.1064371 1.1613333
2.6550877
1.8979895 0.97226685 -2.2151465 -0.8620518 1.5440688 -
1.3822566
-2.6860912 -1.6661612 1.3744977 -1.0391777 -1.7811767 -
2.176058
-2.683843 0.6333658 ]
[ 0.70647556 -0.84163034 -0.61635005 -0.09806145 -0.94011104 -
1.3731344
-1.5828531 -0.7051147 1.7177286 0.6629991 -1.183644
1.365757
1.1996702 0.78505504 -0.45740262 0.72373277 0.6175685
1.1020687
0.87232596 -0.33679235]
[ 2.3379438 0.84163845 1.0935127 -2.6954741 -2.5963457 -
1.3860952
-1.5194715 -3.8016598 1.1649017 2.3203804 -1.8727291
1.743449
0.71060556 1.7861484 -3.975275 3.259786 3.4221208 -
0.28038126
1.1287819 -2.5174115 ]
[-1.3786786 4.593809 4.017309 1.5733426 1.4282767
3.8676436
1.0845753 0.7656716 -0.89666516 -0.18480739 1.4078115 -
1.3665434
-4.5938864 -0.55633205 2.5239277 -1.9437135 -2.048711 -
4.001604
-3.381099 2.3789213 ]
[-0.15464243 0.11477477 -0.23168097 0.40927216 -0.05534938
0.9787869
-0.43060255 0.27967963 0.14131376 -0.5265882 -0.0775831 -
0.09416049
-0.5317773 0.09114048 -0.04401767 0.01090568 0.18210857
0.02474426
-0.43501747 0.11263337]]
Biases_1 :
[-0.27816528 0.07537668 0.2235545 -0.034223 0.09656958 -
0.33741322
0.13071994 -0.14964435 -0.35520387 -0.09051806 0.12850273 -
0.07179476

```

```

    0.2803303 -0.00116276 -0.06033725 -0.04076784 -0.28170696 -
0.05536415
    0.22315164 0.04844811]
Weights_2 shape: (20, 4)
Biases_2 shape: (4,)
Weights_2:
[[-0.8887355 -2.2249074 -0.20965238 2.0001254 ]
 [ 2.170734 -2.1416442 -0.66643167 0.14397201]
 [ 1.3663088 -1.2122456 -0.7336468 0.57629997]
 [ 1.0599803 0.6621496 0.88336825 -1.4989014 ]
 [ 0.942763 1.3304884 0.3388156 -2.215885 ]
 [ 3.0550513 -1.551918 -1.2341331 -0.85893375]
 [ 1.1198447 1.7049388 -1.1786848 -1.5933833 ]
 [ 0.4015014 0.80214363 1.3311552 -2.0633595 ]
 [-1.5480622 -2.8081794 0.56930155 1.18402 ]
 [-0.37867972 -1.2749367 -1.0607828 1.5291333 ]
 [ 1.2594825 1.902686 -0.02482418 -1.8024917 ]
 [-1.4518723 -2.8072965 0.4270706 1.0202515 ]
 [-3.0916548 1.6910958 0.12757272 -0.0650749 ]
 [-1.4351814 -4.189594 0.03743736 0.99723387]
 [ 1.0376905 0.5810701 1.15882 -2.327741 ]
 [-0.3676645 -1.5954238 -1.232485 2.572134 ]
 [-1.3306599 -2.27753 -1.5205495 1.2905151 ]
 [-2.4652572 1.5863322 0.35481012 -0.34256873]
 [-3.2685227 0.5758436 0.42881155 0.26130295]
 [ 1.2171518 0.6978803 0.39253807 -1.5273511 ]]
Biases_2 :
[ 0.16399015 -0.02501617 -0.13355961 -0.0101151 ]

```

Save the model

```

model_filename = 'BQCmodelSoftmaxTF'+tf.__version__+'.keras'

model.save(model_filename)
print(f'Model saved as {model_filename}')

print(f'End of program... ')

Model saved as BQCmodelSoftmaxTF2.21.0.keras
End of program...

```